INCH - POUND
MIL-S-83505/6D
25 September 1991
SUPERSEDING
MIL-S-83505/6C
30 November 1987

MILITARY SPECIFICATION SHEET

SOCKETS (LEAD, ELECTRONIC COMPONENTS) (TYPE VI, SOLDERLESS SPRING CONTACT)

This specification is approved for use by all Departments and Agencies of the department of Defense.

The requirements for acquiring the sockets described herein shall consist of this specification and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-83505

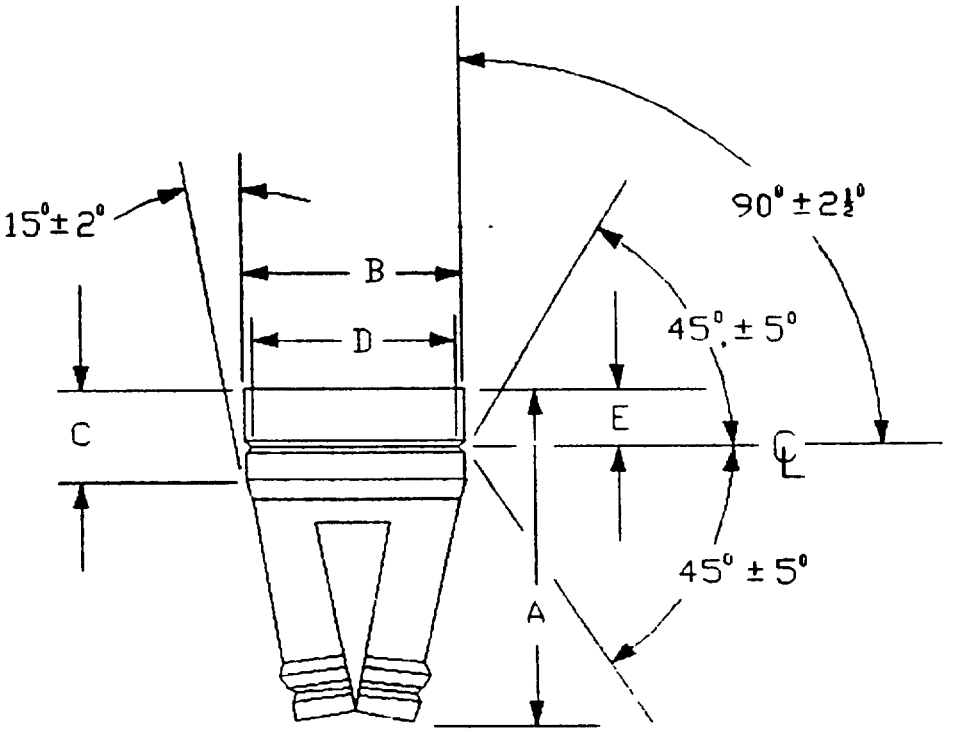


FIGURE 1. Socket, lead, (type VI) solderless spring contact.

Ⓓ denotes changes

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Dash number	Accepts pin		A	B Dia	C	D	E
	Length	Dia	±.003	±.0005	±.003	±.003	±.003
-001	.100 min	.021 max .016 min	.094	.044	.020	.040	.014
-002	.140 min	.030 max .021 min	.136	.0625	.025	.054	.014
-003 see note 1	.140 min	.035 max .030 min see note 2	.136	.0625	.025	.054	.014
-004	.160 min	.044 max .038 min see note 3	.156	.086	.024	.077	.014

Dash number	Recommended mounting hole dia. ±.002 see note 4	Test length (minimum)	Test gauge mating force (max) (lbs)
-001	.041	.100	2.0
-002	.057	.140	3.5
-003	.057	.140	3.5
-004	.081	.160	4.5

FIGURE 1. Sockets (type VI) solderless spring contact - Continued.

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NOTES:
1. Also accepts .025 square post.
2. Use gauge pin with a diameter of.0300 (0.76 mm) +.0002 (0.005 mm), -.0002 minimum and .0350 (0.889 mm) +.0000, -.0002 maximum.
3. Use gauge pin with a diameter of .0380 (0.965 mm) +.0002, -.0000 minimum and .0440 (1.118 mm) +.0000, -.0002 maximum.
4. Plated through hole in accordance with MIL-STD-275. For test purposes, plated through holes shall be at upper limit area .041 (1.041 mm) to .043 (1.092 mm) for -011, .057 (1.448 mm) to .059 (1.499 mm) for -002, -003, and .081 (2.057 mm) to .083 (2.108 mm) for -004.

Inches	mm	Inches	mm	Inches	mm	Inches	mm
.001	0.03	.050	1.27	.084	2.13	.170	4.32
.010	0.25	.052	1.312	.086	2.18	.175	4.44
.016	0.41	.053	1.35	.093	2.36	.185	4.70
.017	0.43	.054	1.37	.098	2.49	.187	4.75
.018	0.46	.055	1.40	.100	2.54	.203	5.16
.020	0.51	.060	1.52	.156	2.64	.219	5.56
.021	0.53	.061	1.55	.110	2.79	.232	5.89
.028	0.71	.062	1.57	.115	2.92	.250	6.35
.030	0.76	.068	1.73	.116	2.95	.281	7.14
.031	0.79	.070	1.78	.120	3.05	.331	8.41
.044	1.12	.078	1.98	.140	3.56	.332	8.43
.047	1.19	.082	.2.08	.155	3.94		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.
4. Unless otherwise specified, tolerance is ± .005 (0.13 mm).

FIGURE 1. Socket, lead, (type VI) solderless spring contact - Continued.

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REQUIREMENTS:

Dimensions and configurations. See figure 1.

Terminals: Type VI (solderless spring contact).

Mounting: Use .046 inch (1.16 mm) minimum thickness mounting board containing plated through holes.

Test mounting: Foil clad laminate in accordance with MIL-P-13949/3.

Contact resistance: .015 ohm initial; .030 ohm after conditioning between test gauge pin and plated through hole.

Spring contact retention: A 4-pound axial load shall be used. The test shall be performed with the socket mounted in a test board. Any separation or loosening of the contact from the test board shall be an indication of noncompliance.

Terminal strength: Not applicable.

Unmating force (for -004): 1 ounce minimum.

Resistance to soldering heat: MIL-STD-202, method 210, except the mounting board soldered with components in one-half of the board and the other half unprotected be soldered using the solder wave technique. Solder in sockets shall not be a failure.

Group qualification: All dash numbers require all qualification tests of MIL-S-83505. When qualifying all dash numbers at once, only six samples of each are required for solderability and resistance to soldering heat.

(D) Part or Identifying Number (PIN): M83505/6 - (dash number from figure 1).

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 17, 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5935-3795-03)

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